CASE STUDY 12

ATTENDANCE RECOVERY COMPUTATION

Python Program:

```
print("ATTENDANCE RECOVERY SYSTEM")
print("-" * 35)

total_classes = int(input("Enter total classes conducted: "))
attended_classes = int(input("Enter classes attended: "))
required_percentage = float(
    input("Enter required attendance percentage: ")
)

# Validate inputs
if total_classes <= 0:
    print("Error: Total classes must be greater than zero.")

elif attended_classes < 0 or attended_classes > total_classes:
    print("Error: Invalid number of attended classes.")

elif required_percentage <= 0 or required_percentage > 100:
    print("Error: Required percentage must be between 0 and 100.")

else:
    current_percentage = (
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    attended_classes / total_classes
) * 100

print("\nCurrent attendance: {:.2f}%".format(
    current_percentage
))

if current_percentage >= required_percentage:
    print("Status: Attendance requirement satisfied.")
    print("Additional classes required: 0")

else:
    print("Status: Attendance requirement not satisfied.")

# Future attendance scenarios
scenarios = [100, 90, 80]

print("\nATTENDANCE RECOVERY ANALYSIS")
print("-" * 35)

for future_rate in scenarios:
    future_classes = 0
    recovery_possible = False

    # Check up to 100000 future classes
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while future_classes <= 100000:

    future_attended = int(
        future_classes * future_rate / 100
    )

    total_after = total_classes + future_classes
    attended_after = attended_classes + future_attended

    new_percentage = (
        attended_after / total_after
    ) * 100

    if new_percentage >= required_percentage:
        recovery_possible = True
        break

    future_classes += 1

if recovery_possible:
    print(
        "Future attendance {}%: {} additional classes".format(
            future_rate, future_classes
        )
    )
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            print(
                "Projected attendance: {:.2f}%".format(
                    new_percentage
                )
            )

        else:
            print(
                "Future attendance {}%: "
                "Requirement cannot be reached "
                "within the search limit.".format(future_rate)
            )

print("\nProgram completed.")
```

Output:

